

In [ ]: It **is** a sequence of characters that forms a search patterns.  
It **is** used **for** search pattern to find a string **or** **set** of string.

Input>>text **format**>> string datatype

re>>Regular Expression

re>>RegEx>>regex

**import** re

In [1]: **import** re

## We want to access 100 pan card data

In [ ]: image 1 dob pan no.  
image 2 dob pan no.  
.  
.  
.  
.  
.  
image n dob pan no.  
  
Image

In [ ]: Image **or** pdf **format**  
PDF:-1.readable  
2.Non readable(image)

1. Readable:

- 1)pdftotext
- 2)PyPdf2
- 3)PDFNumber
- 4)Tabula-py
- 5)camelet

2.Non Readable Pdf

- 1)PDF to image-pdf2image
- 2)Image to text
- 3)pytesseract(open source)
- 4)Google vision
- 5)Amazon Textract
- 6)Easy-OCR

## Regular expression Functions

```
In [ ]: 1)findall()
        2)sub()
        3)match()
        4)search()
        5)compile()
        6)split()
```

### 1)findall()

```
In [ ]: Syntax: re.findall(pattern,text)
        This function help us to find all pattern occurrences in a string.
        It will return list of all matches.

        pattern>>str
        text>>str
```

```
In [3]: import re
        text="""My mobile numbers are:
        9889675656 and 8899776655 and
        9970981730 and 895623784512 Todays date is 20-08-2026"""
        mob_num=re.findall(r"\b[0-9]{10}\b",text)
        mob_num
```

```
Out[3]: ['9889675656', '8899776655', '9970981730']
```

### Access Date from data

```
In [4]: import re
        text="""My mobile numbers are 9889675656 and 8899776655 and
        9970981730 and 895623784512 Todays date is 20-08-2026
        24-10-2026
        12/10/2026
        28/11/2026"""
        date=re.findall("[0-9]{2}[-/][0-9]{2}[-/][0-9]{4}",text)
        date
```

```
Out[4]: ['20-08-2026', '24-10-2026', '12/10/2026', '28/11/2026']
```

## Access Pan card data

```
In [5]: import re
text="""Pan Card Numbers :
ASDFG1234W
POIUY3456P
JHGFD3456U
KLJHG5678I
DFGH678JH8"""

pattern="[A-Z]{5}[0-9]{4}[A-Z]{1}"
pan_card=re.findall(pattern,text)
pan_card
```

```
Out[5]: ['ASDFG1234W', 'POIUY3456P', 'JHGFD3456U', 'KLJHG5678I']
```

## Access Adhar Card number

```
In [6]: import re
text="""Aadhar card numbers:
2354 4569 7895
6895 4562 1236
4561 7854 1256
"""

a_num=re.findall("[0-9]{4}\s[0-9]{4}\s[0-9]{4}",text)
a_num
```

```
Out[6]: ['2354 4569 7895', '6895 4562 1236', '4561 7854 1256']
```

## Access IFSC codes

```
In [7]: import re
text="""Bank IFSC codes:
HDFC0000182
SBIN0000345
ICIC9078675"""

pattern="[A-Z]{4}[0-9]{7}"
ifsc_code=re.findall(pattern,text)
ifsc_code
```

```
Out[7]: ['HDFC0000182', 'SBIN0000345', 'ICIC9078675']
```

```
In [8]: import re
text="""Email Id
bhagyashri@gmail.com
chaitnya@yahoo.com
yashrakte12@rediffmail.com
"""

pattern="[a-z][0-9][@]{1}[a-z][.com]"
email_id=re.findall(pattern,text)
email_id
```

```
Out[8]: []
```

```
In [29]: import re
text="""Email Id
bhagyashri@gmail.com
chaitnya@yahoo.com
yashrakte12@rediffmail.com
"""

email_id=re.findall(r"[A-Za-z0-9._]+@[A-Za-z.-]+\.[a-zA-Z]{2,}",text)
email_id
```

```
Out[29]: ['bhagyashri@gmail.com', 'chaitnya@yahoo.com', 'yashrakte12@rediffmail.com']
```

```
In [28]: import re

text = """Email Id
bhagyashri@gmail.com
chaitnya@yahoo.com
yashrakte12@rediffmail.com
"""

email_id = re.findall(
    r"[A-Za-z0-9._]+@[A-Za-z.-]+\.[a-zA-Z]{2,}",
    text
)

print(email_id)
```

```
['bhagyashri@gmail.com', 'chaitnya@yahoo.com', 'yashrakte12@rediffmail.com']
```

## Speacial Sequences

### 1.\d:String contain digit

```
In [30]: import re
text="""Aadhar card numbers:
2354 4569 7895
6895 4562 1236
4561 7854 1256
"""

a_num=re.findall("\d{4}\s\d{4}\s\d{4}",text)
a_num
```

```
Out[30]: ['2354 4569 7895', '6895 4562 1236', '4561 7854 1256']
```

### 2.\D

```
In [ ]: Returns a match where the string does not contain digit
```

```
In [31]: import re
text="abc1234python_23"
result=re.findall("\D+",text)
result
```

```
Out[31]: ['abc', 'python_']
```

### 3.\w

```
In [ ]: It returns a match the string contains any word characters
(characters for [a-z][0-9][_])
```

```
In [32]: import re
text=text = """Email Id
bhagyashri@gmail.com
chaitnya@yahoo.com
yashrakte12@rediffmail.com
"""
mail_id=re.findall("\w{7,30}@[a-z.]{5,15}",text)
mail_id
```

```
Out[32]: ['bhagyashri@gmail.com', 'chaitnya@yahoo.com', 'yashrakte12@rediffmail.com']
```

### 4.\W

```
In [ ]: \w>>Except[a-zA-Z0-9_]
\W>>[^a-zA-Z0-9_]
```

```
Out[2]: [ '\n',
```

## 5.\s

```
In [3]: import re
text="""Aadhar card numbers:
2354 4569 7895
6895 4562 1236
4561 7854 1256
"""
a_num=re.findall("\d{4}\s\d{4}\s\d{4}",text)
a_num
```

```
Out[3]: ['2354 4569 7895', '6895 4562 1236', '4561 7854 1256']
```

## 6.\S

```
In [ ]: \S>>Except Space
```

```
In [6]: import re
text="""Aadhar card numbers:
2354 4569 7895
6895 4562 1236
4561 7854 1256
"""
a_num=re.findall("\S{5}",text)
a_num
```

Out[6]: ['Aadha', 'numbe']

## 7.\b

```
In [7]: import re
text="""Aadhar card numbers:
2354 4569 7895
6895 4562 1236
4561 7854 1256
"""
a_num=re.findall("\d{4}\s\d{4}\s\d{4}",text)
a_num
```

Out[7]: ['2354 4569 7895', '6895 4562 1236', '4561 7854 1256']

```
In [12]: import re
text="""Aadhar card numbers:
2354 4569 789534#invalid
6895 4562 1236#valid
4561 7854 125689
"""
a_num=re.findall(r"\b\d{4}\s\d{4}\s\d{4}\b",text)
a_num
```

Out[12]: ['6895 4562 1236']

## Pytesseract

In [ ]: It **is** a optical character recongnition(OCR) tool **for** python  
It will read **and** recognize the text **from** image

```
In [14]: import pytesseract
```

```
In [15]: pip install pytesseract
```

Requirement already satisfied: pytesseract in c:\users\sai\anaconda3\anaconda\lib\site-packages (0.3.13)  
Requirement already satisfied: packaging>=21.3 in c:\users\sai\anaconda3\anaconda\lib\site-packages (from pytesseract) (23.1)  
Requirement already satisfied: Pillow>=8.0.0 in c:\users\sai\anaconda3\anaconda\lib\site-packages (from pytesseract) (11.0.0)  
Note: you may need to restart the kernel to use updated packages.

```
In [ ]: How to install pytesseract
step 1-click on link https://github.com/UB-Mannheim/tesseract/wiki
step 2-tesseract-ocr-w64-setup-5.5.0.20241111.exe (64 bit)
check bit version and download it.
step 3-Install pytesseract
step 4-check your tessract path in c drive programs file(both)
step 5-copy that Tessract-OCR and tessract.exe path
step 6-set path in your environment variable.
```

```
In [16]: import pytesseract
pytesseract.pytesseract.tesseract_cmd=r"C:\Program Files\Tesseract-OCR\tessera
pytesseract.pytesseract.tesseract_cmd
```

```
Out[16]: 'C:\\Program Files\\Tesseract-OCR\\tesseract.exe'
```

```
In [17]: from PIL import Image
import pytesseract

img = Image.open(r"H:\bhagashridiske\All my documents\Adhar card.jpeg")
text = pytesseract.image_to_string(img)
print(text)
```

```
aval aaarey area
> Bhagyashri Navnath Ghaime
9) ae AY / Year of Birth : 1994
```

```
' ) €41/ Female
```

```
7530 9817 9936
ee | 4G
```

```
arte Fea er
```



```
In [20]: def get_pan_details(image_path):
pan_data=pytesseract.image_to_string(image_path)

dob=re.findall("\d{2}[-/]\d{2}[-/]\d{4}",pan_data)
if dob:
    date_of_birth=dob[0]
else:
    date_of_birth=""
print(pan_data)
print("Date of Birth:",date_of_birth)
get_pan_details('pan_card3.jpg')
```

| INCOME TAX DEPARTMENT 2 GOVT. OF INDIA

cart een cen ae  
Permanent Account Number Card  
QGBPK8700C

are  
ara / Name  
KOCHERLA SRIKANTH

fire ait aT / Father's Name  
MUKKANTI KOCHERLA

'am it Arete Date of Birth K Si komth.  
04/05/1997 aa ieeetire

Date of Birth: 04/05/1997

## 2.re.sub

In [ ]: Used to replace one **or** many matches.  
  
Syntax: re.sub(old\_substring\_pattern,new\_substring\_pattern,text)  
  
Return updated string

```
In [23]: string1="python and data science 2025,,2026 "  
string2=string1.replace('2025','').replace('2026','').replace(',','').strip()  
string2
```

Out[23]: 'python and data science'

```
In [24]: string1="python and data science 2025,,2026 "  
new_string=re.sub('\d','*',string1)  
new_string
```

Out[24]: 'python and data science \*\*\*\*,,\*\*\*\* '

```
In [25]: string1="python and data science 2025,,2026 "  
new_string=re.sub('\d','',string1)  
new_string
```

```
Out[25]: 'python and data science ,, '
```

```
In [26]: string1="python and data science 2025,,2026 "  
new_string=re.sub('[0-9,]', '',string1)  
new_string
```

```
Out[26]: 'python and data science '
```

```
In [28]: string1="python and data science 2025,,2026 "  
new_string=re.sub('[0-9,]', '',string1).strip()  
new_string
```

```
Out[28]: 'python and data science'
```

```
In [33]: string1="python    and    data    science "  
new_string=re.sub('\s{1,25}',' ',string1)  
new_string
```

```
Out[33]: 'python and data science '
```

```
In [34]: string1="123 python    and    data    science#$$ 2024 *&^%@"  
result=re.findall('[a-zA-Z]',string1)  
result
```

```
Out[34]: ['p',  
          'y',  
          't',  
          'h',  
          'o',  
          'n',  
          'a',  
          'n',  
          'd',  
          'd',  
          'a',  
          't',  
          'a',  
          's',  
          'c',  
          'i',  
          'e',  
          'n',  
          'c',  
          'e']
```

## 1.Plus(+)

**+>>**One **or** more occurrences

```
string="123 python and data science#$$ 2024 *&^%@"
result=re.findall('[^a-zA-Z]{1,20}',string)
result
```

```
[ '123 ', ' ', ' ', ' ', ' ', '#$% 2024 *&^%@ ' ]
```

```
string="123 python and data science#$$ 2024 *&^%@"
result=re.findall('[^a-zA-Z]+',string)
result
```

```
[ '123 ', ' ', ' ', ' ', ' ', '#$% 2024 *&^%@ ' ]
```

```
string="123 python and data science#$$ 2024 *&^%@"
result=re.sub('[^a-zA-Z]+'," ",string)
result
```

```
' python and data science '
```